

Experiment No-03

Aim: Write an Assembly language program (ALP) to find ascending order of ten numbers in memory and store the result into memory.

Software Used: Keil uVision5

Program:

```
                AREA file3, CODE, READONLY
                EXPORT __main
__main
                MOVs R4, #9

OUTER           MOVs R0, #9
                LDR R1, =0x1FFFF000

INNER          LDRB R2, [R1]
                LDRB R3, [R1, #1]

                CMP R2, R3
                BLS SKIP

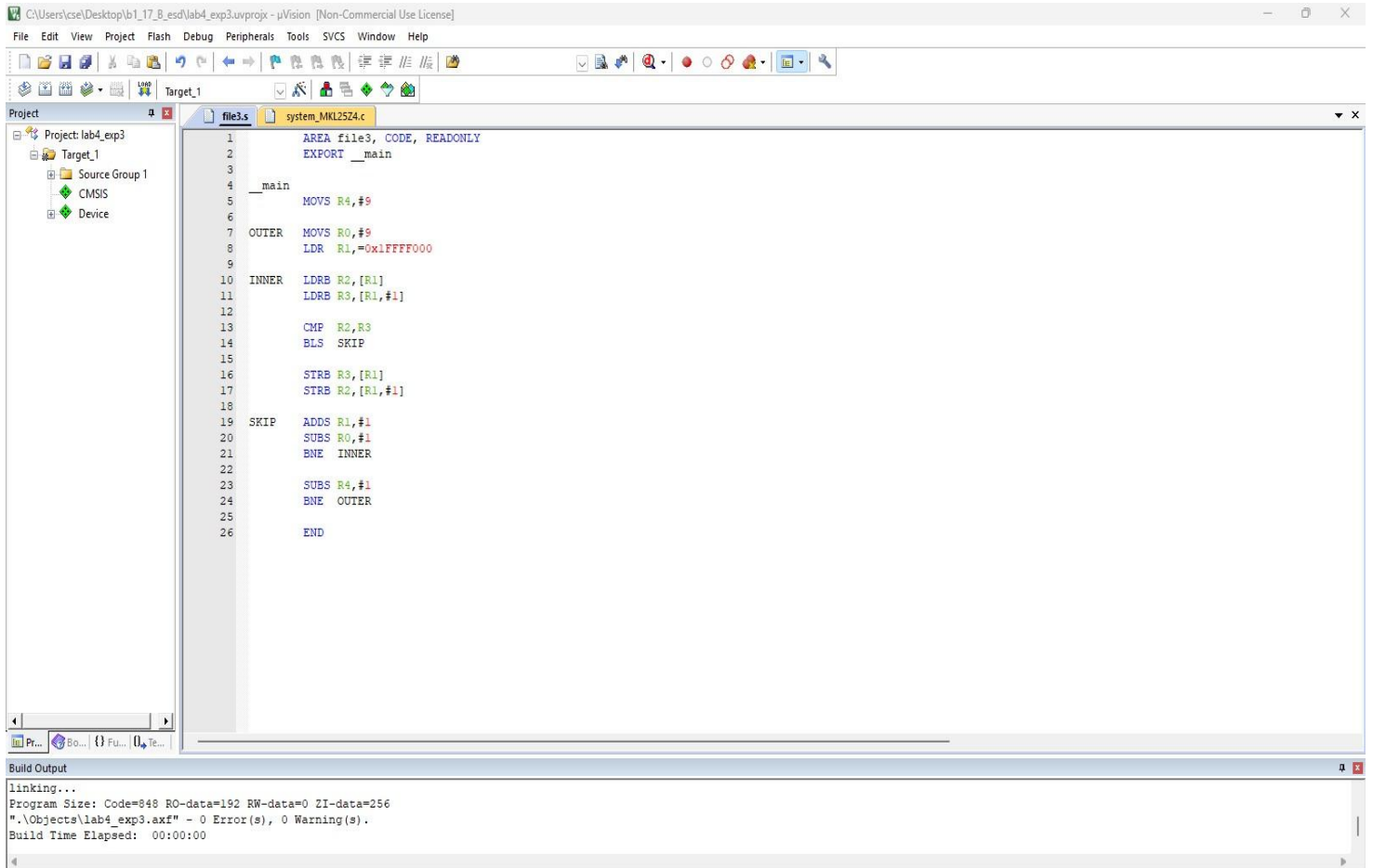
                STRB R3, [R1]
                STRB R2, [R1, #1]

SKIP           ADDS R1, #1
                SUBS R0, #1
                BNE INNER

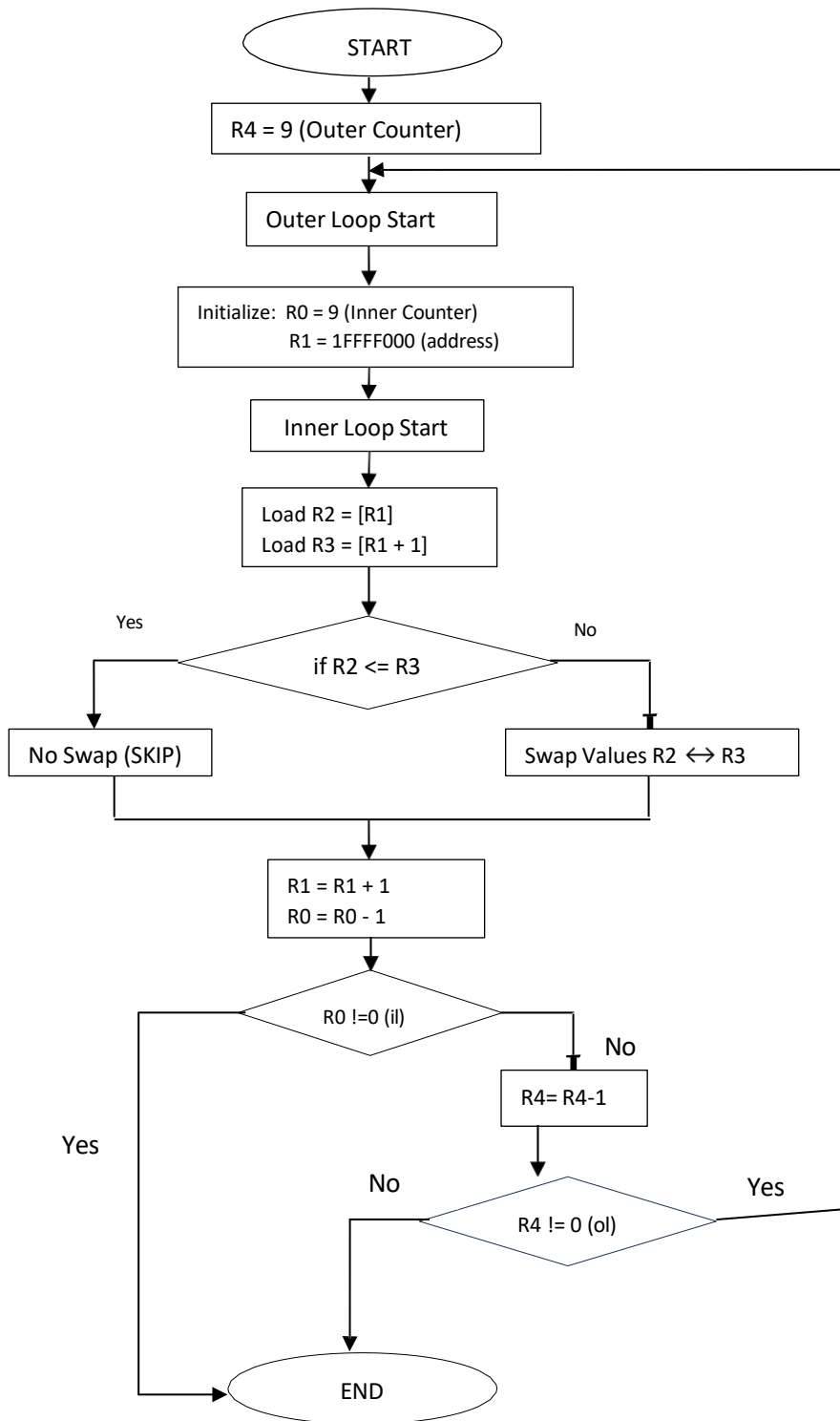
                SUBS R4, #1
                BNE OUTER

                END
```

Program Screenshot:

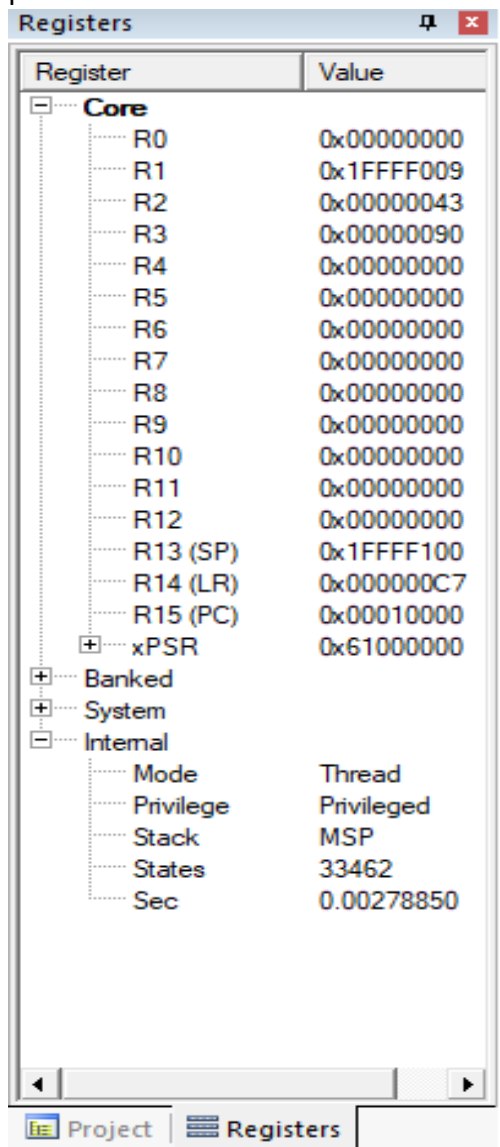


Flowchart:



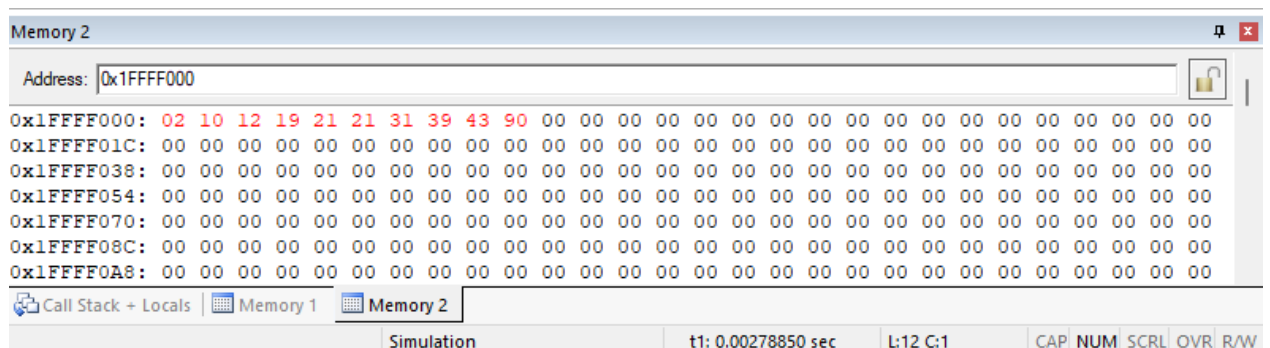
Results and Conclusion:

Output:



Register	Value
Core	
R0	0x00000000
R1	0x1FFFF009
R2	0x00000043
R3	0x00000090
R4	0x00000000
R5	0x00000000
R6	0x00000000
R7	0x00000000
R8	0x00000000
R9	0x00000000
R10	0x00000000
R11	0x00000000
R12	0x00000000
R13 (SP)	0x1FFFF100
R14 (LR)	0x000000C7
R15 (PC)	0x00010000
xPSR	0x61000000
Banked	
System	
Internal	
Mode	Thread
Privilege	Privileged
Stack	MSP
States	33462
Sec	0.00278850

Memory:



Address	0x1FFFF000
0x1FFFF000:	02 10 12 19 21 21 31 39 43 90 00
0x1FFFF01C:	00 00
0x1FFFF038:	00 00
0x1FFFF054:	00 00
0x1FFFF070:	00 00
0x1FFFF08C:	00 00
0x1FFFF0A8:	00 00

10 Numbers are arranged in ascending order and is available in Register R1 and Memory